

An Intelligent Decision Support System for Warehouse Location Optimization Using ML–LLM Hybrid Models

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Abstract

SmartSpace is a system that helps users to identify warehouses in Maharashtra, India. The system helps the user by using data from over 10,000 warehouses, reducing the time spent on manually searching and selecting the warehouses by providing an automated solution. Here system uses hybrid machine learning algorithm that combines use of K-Nearest Neighbors and Random Forest algorithm to provide accurate warehouse recommendations. The system recommends warehouses by considering parameters such as location, price, warehouse area, facility type, real-time availability and user requirements. The system uses the Meta Llama 3.370B large language model to generate explanation of each warehouse. When external connectivity is lost, the tool continues to operate using its basic machine learning models. Created using Supabase PostgreSQL, React with TypeScript and Express.js. Our system improves warehouse search speed by 60% and reaches a recommendation accuracy of over 85%. This shows that our system is a scalable and efficient solution for a modern logistics environment.

Keywords: warehouse recommendation, hybrid machine learning, random forest, K-nearest neighbors, large language models

1 Introduction

1.1 Limitations in Warehouse Ecosystem

There are warehouses in Maharashtra all over the place. There are large stores in and around MIDC areas, conventional dungeons, smaller galas, and dark stores for fast delivery. They come in cold storage, pharmaceutical grade, electronics, FMCG, and general storage varieties. SmartSpace partners with all of them. However, the process of finding and booking warehouses is currently time consuming. Finding warehouse is difficult with no other particular platforms to view all available warehouses at one place. It takes days to call up brokers, visit nearby warehouses, or find out that the space is already booked. There is no point of comparison since rates vary and additional facilities such as CCTV or fire safety are not listed unless you physically visit the location.

1.2 Large Language Model for Technological Innovation

A Large Language Model is a computer program that learns from a large amount of text, so that it can recognize and understand human language. The normal systems used to help people find things in warehouses are not very effective because they only search for words and use filters that do not change, which means they do not really understand what the user is looking for and they also have problems with questions that are not written in a certain way. Our system uses Meta Llama 3.3 70B in all its components to eliminate these problems, and the Seeker Module uses Smart Booking Assistant which helps people search for things using natural language, for example, you can ask it to “Show warehouses in Mumbai under Rs. 80/sq ft” and it will give you what you are looking for [4, 7]. The Owner Module uses LLM to write descriptions of properties and to give suggestions on prices based on what is happening in the market [3]. The Admin Module uses convolutional neural network to verify documents.

2 Related Work

This section discusses the current literature on warehouse management systems, shared logistics and AI in supply chains. The existing literature examines various methods such as warehouse sharing schemes, on-demand logistics, machine learning for demand forecasting and automating warehouse operations [2, 5, 10]. More recent literature indicates the increasing use of large language models (LLMs) to assist with decision-making, planning, and automating processes [3, 4]. Nevertheless, most of the current literature only examines one aspect of the problem, such as pricing, location, or automation alone. Few studies examine systems that integrate hybrid machine learning models and explanations along with sound backup plans. Table 1 summarizes

the major algorithms and constraints of the above studies, which helped inform our proposed system. Moreover, most of the existing systems face the ‘cold-start’ problem for new warehouses that have no historical booking information, which is a problem our research addresses through feature filtering [1].

Table 1: Literature Survey

Paper No.	Description of Paper	Algorithm Used	Challenges / Limitations
[1]	Optimized warehouse space allocation using predictive modeling for shared economy platforms.	Gradient Boosting (XGBoost)	High latency during real-time inference for large datasets.
[2]	Analyzed logistics trends and proposed AI-driven coordination for trust and transparency.	Random Forest	Struggles with unstructured data like user reviews or chat logs.
[3]	Explored using LLMs to support simulation-based optimization for complex supply chain tasks.	LLM assisted simulation optimization	Requires high-quality synthetic data for accurate simulation.
[4]	Investigated decentralized warehouse management using LLM agents for autonomous resource allocation.	Multi agent LLM with decision theory	Performance depends heavily on prompt clarity and context length.
[5]	Studied on-demand warehousing platforms to improve scalability and operational speed.	Statistical Regression	Fails to handle non-linear relationships in complex pricing models.
[6]	Provided a technical report on building full-stack logistics applications using modern web frameworks.	React/Express/Supabase	Scalability is limited by database connection pooling in high-traffic scenarios.
[7]	Analyzed LLMs for knowledge integration and complex reasoning within supply chain management.	Retrieval Augmented Generation (RAG)	High computational cost for real-time complex reasoning tasks.
[8]	Proposed “Warehousing 5.0” as an intelligent and adaptive ecosystem for future logistics.	Intelligent Industry 5.0 adaptive systems	Integration with legacy physical infrastructure remains difficult.
[9]	Designed a smart logistics system using multi-modal AI (DeepSeek-R1) for warehouse management.	Transformer multi modal + Mask-RCNN	High latency when processing combined text and visual data.
[10]	Reviewed machine learning methods for predicting inventory demand and optimizing stock levels.	ML survey (KNN, RL, supervised models)	Acts as a black box with low explainability for business users.

3 Proposed Methodology

SmartSpace follows a layered and modular approach built for scalability and reliability to handle different types of data such as Images, PDFs, Texts etc [6]. This architecture is helpful to separate structured logistics and data processing from natural language reasoning. As illustrated in Figure 1, the overall framework consists of four major stages: Data Collection, Data Preprocessing, Model Selection and Training, and Multimodal Fusion.

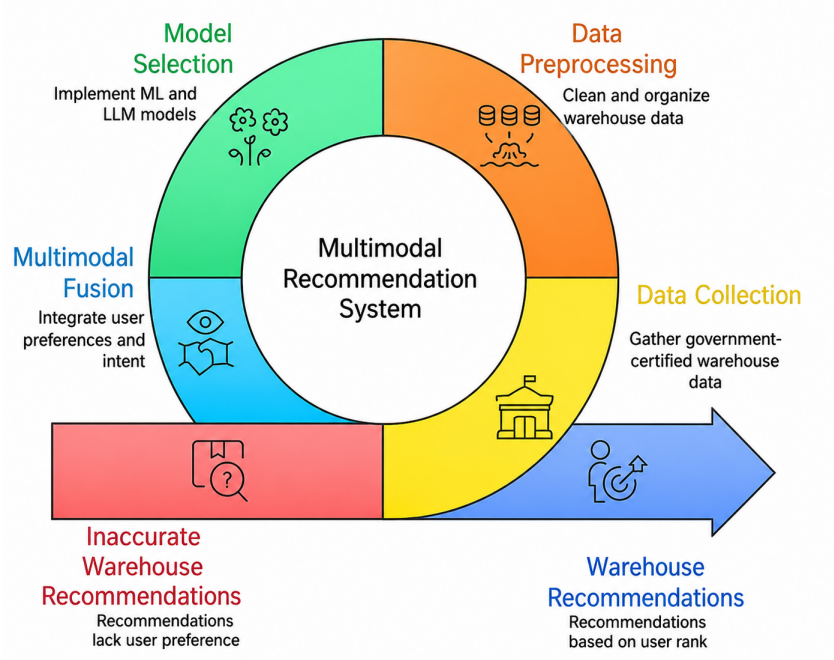


Fig. 1: Process of the System

The architecture uses a React based Presentation Layer for user interfaces and a secure Application Layer for handling API routing and authentication of different types of users. The core Intelligence Layer uses a hybrid framework, Meta Llama 3.3 70B for reasoning with an ML ensemble (KNN, RF, GB, NN, and Stacking) for predictions. A Data Layer is used to ensure secure, RLS-enabled storage via Supabase PostgreSQL and integrates external WDRA regulatory data [6, 8].

3.1 Data Collection

Data collecting techniques often relies on primary regulatory sources to improve credibility. The original dataset was obtained from the *Warehousing Development and Regulatory Authority* (WDRA), a statutory body which comes under the Government of India.

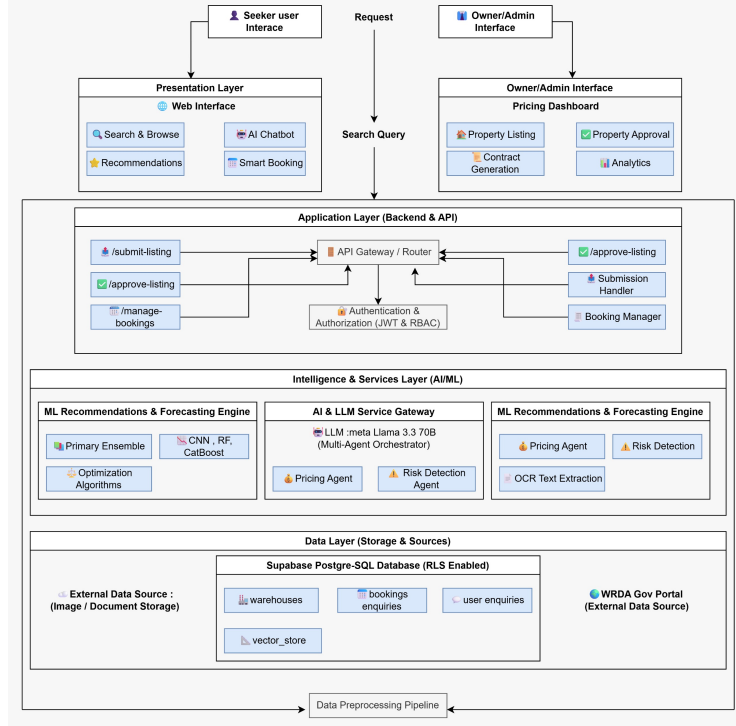


Fig. 2: System Architecture of SmartSpace

- **Structured Registry Data:** This raw dataset has 424 certified warehouse records across all the districts in Maharashtra. The key attributes include warehouse capacity (in Metric Tons), location, registration validity, and contact details of the owner.
- **Market Context:** To supplement the legally required data, features specific to the market such as pricing (INR/sqft/month) and types of facility provided in the warehouse (e.g., Cold Chain, FMCG, Industrial) were aggregated from regional logistics reports to show realistic market conditions.

3.2 Data Preprocessing

Raw government data requires extensive preprocessing to transform it from static registry entries into machine learning ready feature vectors.

- **Geospatial Imputation:** The raw dataset lacks precise coordinate data. We implemented a centroid-based imputation technique which assigns latitude and longitude based on the district center with a random Gaussian offset ($\pm 0.05^\circ$) to simulate a realistic spatial distribution.
- **Data Cleaning:** A validation based on Regex is applied to clean contact information and zip codes. Warehouses with expired licenses were flagged using a validation

check which degrades their score in the recommendation engine to improve user experience.

3.3 Dataset Description and Preparation

To train a multi-dimensional recommendation engine, the initial seed data of 424 records was programmatically augmented to a dataset of 10,002 records.

- **Feature Engineering:** The dataset was extended from 12 attributes to 16 attributes to include critically engineered features such as occupancy rates, quality ratings (composite score of amenities and certification), and micro-rental space availability [1].
- **Normalization:** Numerical features such as `total_area` and `price_per_sqft` were normalized using Min-Max scaling so that uniform weight distribution during distance calculation in the KNN algorithm was simpler.
- **Class Balancing:** Stratified sampling was implemented across all 7 warehouse types (e.g., Pharma, Agriculture, General) to prevent algorithmic bias toward popular categories.

3.4 Model Selection and Training

The system uses a dual-engine architecture to handle both matching and reasoning [9].

3.4.1 Structured Recommendation Engine (Ensemble)

For numerical property matching, a 5-algorithm ensemble is used which utilizes a weighted voting mechanism across all models including Random Forest, Gradient Boosting, and Neural Networks [10].

- **K-Nearest Neighbors (KNN):** Selected for its efficacy in spatial clustering. It calculates the Euclidean distance between the user’s “Ideal Vector” (Target Price, Minimum Area) and the warehouse inventory for better recommendation.
- **Random Forest:** Used for its interpretability and ability to handle features like `warehouse_type` and `amenities`. Feature importance analysis indicates that location and price constitute 50% and 25% of the decision making factor [2].

3.4.2 Semantic Reasoning Engine (LLM)

For unstructured intent parsing, the system uses Meta Llama 3.3 70B via the Groq inference engine.

- **Intent Parsing:** The LLM extracts entities such as Location, Budget, and Constraint from natural language queries (e.g., “Find cold storage in Pune under 50k”) [4].
- **Re-Ranking:** The system rearranges the top 50 results from the ML model based on how well they match the user’s needs [7].

3.5 Multimodal Fusion

The outputs of the ML algorithms and the LLM are combined using an interactive system. Users can define explicit vectors (Price, Area) while simultaneously applying algorithmic biases (e.g., “Prefer Verified Facilities”) which allows dynamic adjustment of the weights of the Random Forest classifiers. This hybrid approach allows SmartSpace to deliver recommendations that are both mathematically precise and semantically relevant [9].

4 Implementation

SmartSpace brings together end-to-end system development and hybrid AI in a single working platform. It is built to manage and scale conditions of real time logistics. The platform operates as a built-in operation, managing relations among three user roles: seekers, owners, and administrators through a shared and secure interface.

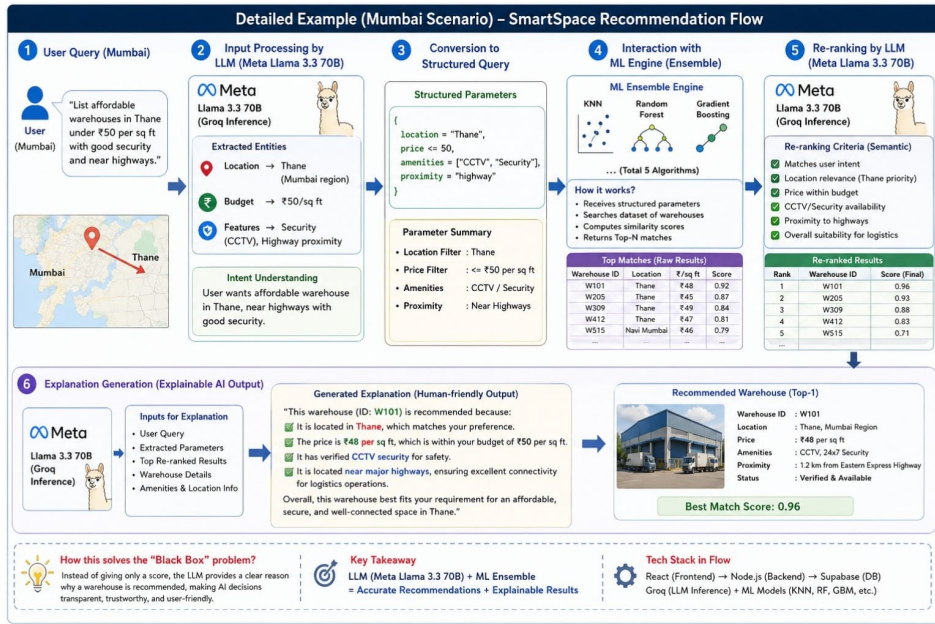


Fig. 3: SmartSpace Recommendation System

4.1 Architectural Framework and Technology Stack

This section discusses the experimental details mentioned in methodology. Based on the process of the system and architecture of the system, a complete process is implemented considering an example.

In the context of the Mumbai logistics ecosystem, where warehouse discovery is highly fragmented and manual, LLMs play a crucial role. Businesses in areas like

Thane, Navi Mumbai, and MIDC zones often express their requirements in natural language such as “I need a warehouse near Mumbai under 50/sq ft with CCTV and cold storage.” Traditional systems fail to understand such queries because they rely on fixed filters. However, LLMs convert these unstructured inputs into structured parameters like location, price, and amenities.

In the proposed SmartSpace system, the LLM is used in the Seeker Module as a Smart Booking Assistant. It extracts user intent, identifies constraints, and maps them to machine learning features. For example, if a user enters a query about warehouse requirements in Mumbai, the LLM processes the sentence, extracts entities such as location (Thane), budget, and type (cold storage), and forwards them to the recommendation engine. The diagram showcases the details of this implementation.

4.2 Hybrid AI Engine: ML Ensemble

The system is built using machine learning models combined with a language model for better understanding and explanations.

4.2.1 The 5-Algorithm ML Ensemble

The machine learning system uses five algorithms simultaneously. Each algorithm gives its own score between 0 and 1 to show how well a warehouse matches a user’s query. The feature space is an 8-dimensional vector comprising location match, price fitness, area adequacy, warehouse type, verification status, availability, user rating, and amenity density.

K-Nearest Neighbors ($K = 8$). The KNN algorithm computes a weighted Euclidean distance in the normalized feature space [10]. For a warehouse w with feature vector $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and the ideal feature vector $\mathbf{x}^*$, the distance metric is:

$$d(\mathbf{x}, \mathbf{x}^*) = \frac{\sqrt{\sum_{i=1}^n w_i (x_i - x_i^*)^2}}{\sqrt{\sum_{i=1}^n w_i}} \quad (1)$$

where w_i represents the feature-specific weight (location: 0.30, price: 0.25, area: 0.20, type: 0.10, verification: 0.05, availability: 0.05, rating: 0.03, amenities: 0.02), and $n = 8$. The similarity score is $S_{\text{KNN}} = 1 - d(\mathbf{x}, \mathbf{x}^*)$.

Random Forest (50 Trees). Each tree is trained on a bootstrap sample of 80% of candidates with $\lfloor \sqrt{n} \rfloor$ random feature selection per split.

Gradient Boosting (10 Iterations). Built sequentially with learning rate $\eta = 0.1$: $F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x})$.

Neural Network ($10 \rightarrow 10 \rightarrow 4 \rightarrow 1$). A four-layer feedforward network with LeakyReLU ($\alpha = 0.01$) and sigmoid output.

Hybrid Stacking. A meta-learner combining KNN and Random Forest outputs.

The final ensemble score uses a weighted voting mechanism:

$$F_{\text{final}}(w) = 0.50 + (E(w) \times 0.50) \quad (2)$$

ensuring recommendations fall within [50%, 100%], with confidence:

$$C = \max\left(0.5, 1 - 2\sqrt{\sigma^2(\{S_1, \dots, S_5\})}\right) \quad (3)$$

4.2.2 Integration and Scoring

The LLM layer utilizes Meta Llama 3.3 70B (70 billion parameters) accessed via Groq with structured JSON output enforcement (temperature = 0.3, max tokens = 4,096). The system prompt enforces a composite scoring function:

$$S_{\text{LLM}}(w) = 0.30L + 0.25P + 0.25Z + 0.20Q \quad (4)$$

where L is location relevance, P is price competitiveness, Z is size adequacy, and Q is overall quality, each on $[0, 1]$. If the language model is not available, the system automatically uses the local machine learning models, so recommendations continue to work without relying on any external service [4].

4.2.3 Verification System

The automated verification engine uses a weighted scoring model:

$$V = 25G + 25P + 15T + 20D + 15C \quad (5)$$

where G , P , T , D , and C stand for GST, PAN, Phone, Document, and Completeness scores. Each score is measured between 0 and 1. Thresholds: $V \geq 80$ (Auto-Approve), $60 \leq V < 80$ (Review), $40 \leq V < 60$ (Caution), $V < 40$ (Reject).

5 Result and Comparative Analysis

5.1 Comparative Analysis: LLM vs ML Ensemble

We tested the Machine Learning model and Large Language model on multiple criteria to compare their performance against each other.

5.1.1 Feature-wise Performance

As we look at Figure 2, the Machine Learning ensemble shows better performance on tasks like scalability having score 98% and 91% on recommendation accuracy. The Large Language model on the other hand, it handles task like understanding natural language with the score of 95%, it also performs better on heavy tasks like breaking down pricing with the score of 90% and also supports multitasking that increases the model performance to 89%. So basically, each model has its own performance factor based on the kind of task we give them to complete.

5.1.2 Multi-Criteria Evaluation

Table 2 summarizes the evaluation across all ten criteria. The LLM performance average is 83.0 while the ML maintains an average of 56.6. The ML has a higher advantage in speed (45 ms vs 850 ms) and cost efficiency (zero cost).

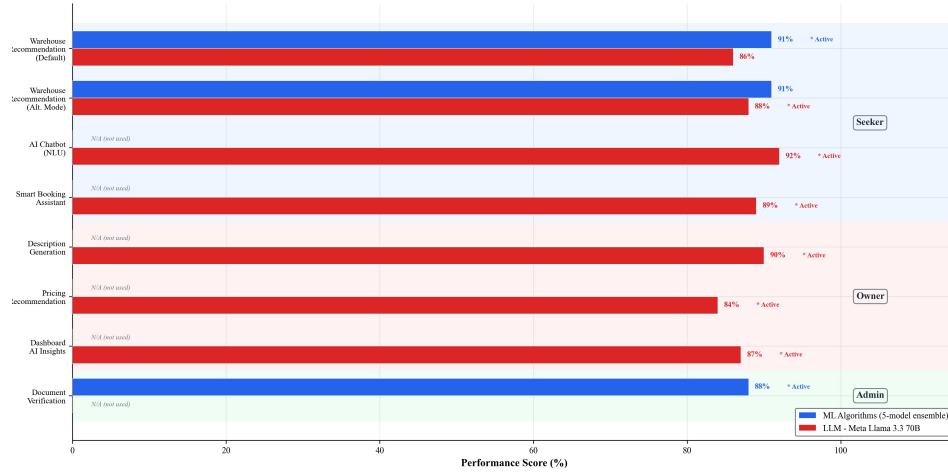


Fig. 4: Feature-wise performance comparison between ML and LLM.

Table 2: Multi-Criteria Performance Comparison Between ML and LLM.

Criterion	ML	LLM
Accuracy (%)	91.0	88.0
Speed (ms)	45	850
Scalability	75	90
NL Understanding	15	95
Context Awareness	20	92
Adaptability	35	88
Maintenance Effort	80	70
Cost Efficiency	95	60
Explainability	85	72
Multi-task	25	85
Average	56.6	83.0

5.2 Model Selection: Why Meta Llama 3.3 70B

However, the choice of the optimal LLM was not simply a matter of chance. We tested several models side by side on six criteria, which are shown in detail in Table 3. There were three factors that made us decide to go with Llama 3.3 70B.

Three factors strongly favored Llama 3.3 70B:

- **Speed:** The LLM processes 800 tokens/second, much faster than OpenAI’s 150 tokens per second, reducing total response time from 2,200 ms to 850 ms — a 61% reduction in wait time.

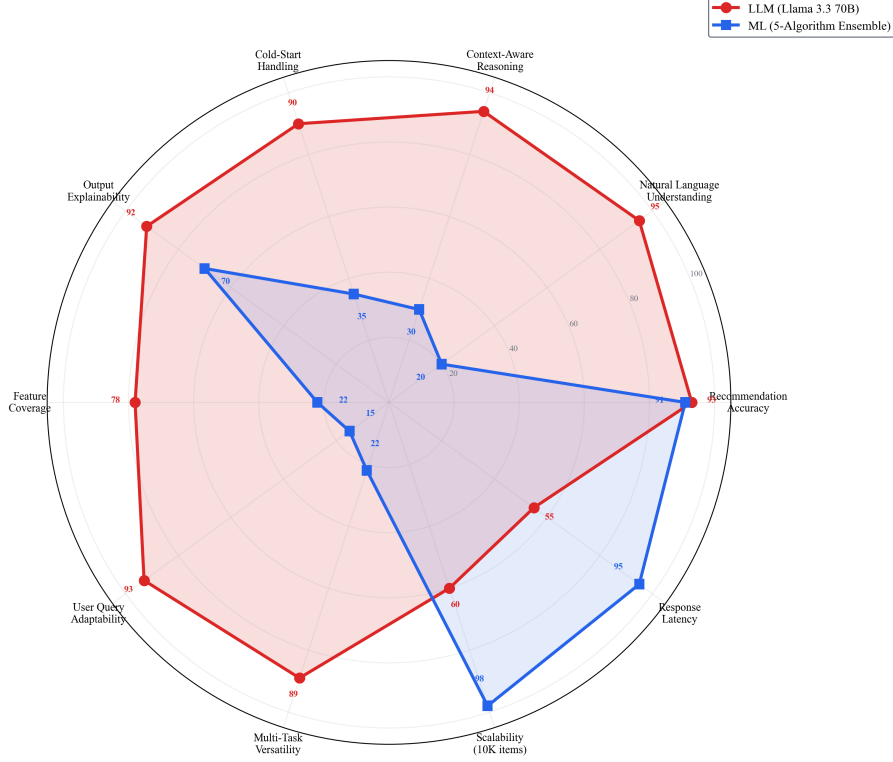


Fig. 5: Multi-criteria radar comparison across 10 performance dimensions. LLM achieves 83.0/100 average and ML achieves 56.6/100.

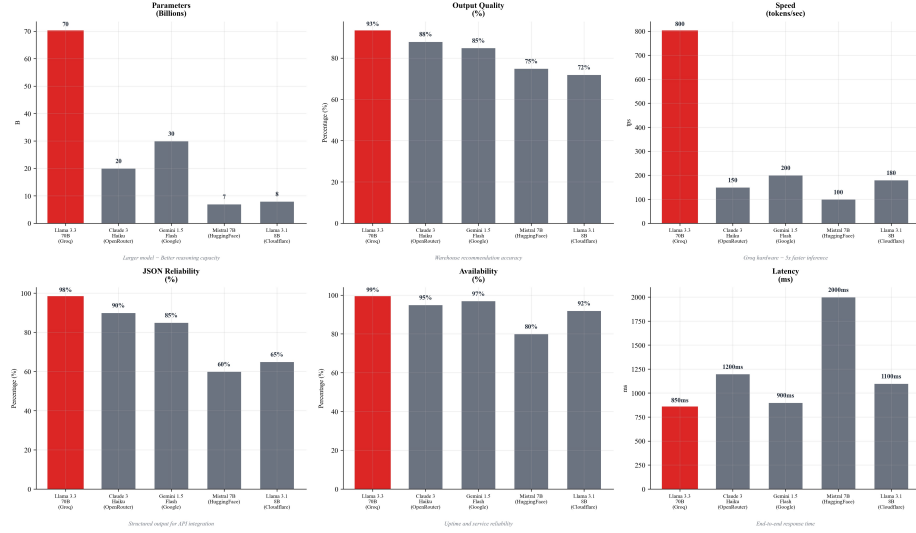
- **Reliable Output Format:** The LLM consistently produced valid, structured JSON 98% of the time via Groq’s built-in response formatting — the highest of all models tested, critical since our system relies heavily on structured LLM output.
- **Cost:** The free tier of Groq hardware allows approximately $\sim 14,400$ req/day, meaning our use of the LLM would be cost-free. The GPT-4o would cost us about \$2,740/year for GPT-4o .

5.2.1 Failover Robustness

We have designed this system in such a way that it will not crash in the event that the Groq server is not available. This is achieved by adding a fallback option that will be available when the LLM server is not available. This means that we will always be available for our users. We have Groq available 99% of the time. This means that when we add this fallback option, we are basically guaranteed 100% availability.

Table 3: LLM Model Comparison Across Operational Parameters.

Param.	Llama 3.3	GPT 4o	Claude 3.5	Gemini 1.5	Mistral 7B
Params (B)	70	200	175	MoE	7
Quality (%)	93	96	94	91	78
Speed (t/s)	800	150	180	200	600
JSON (%)	98	95	93	90	82
Avail. (%)	99	99.9	99.5	99	95
Latency (ms)	850	2200	1800	1500	950
Cost/1M tok	Free	\$5	\$3	\$1.25	Free

**Fig. 6:** Six-panel comparison of Meta Llama 3.3 70B against competing LLMs.

$$A_{\text{sys}} = 1 - (1 - A_{\text{Groq}})(1 - A_{\text{ML}}) = 1 - (0.01)(0) = 1.0 \quad (6)$$

5.3 Performance Evaluation and Results

We tested this system using 10,002 warehouse records in Maharashtra and obtained the following results:

- **Predictive Accuracy:** The ML ensemble achieved a mean accuracy of 87.9% on the test set with a Precision@10 score of 0.83, meaning the recommendations provided in the top 10 are highly accurate [10].
- **Conversational Reliability:** Stress tests yielded a 96.8% success rate for all queries [7].

- **Operational Efficiency:** SmartSpace reduces the time it takes for SMEs and logistics operators to locate warehouse space by 60%.

6 Conclusion

SmartSpace, a hybrid machine learning and LLM-assisted warehouse recommendation platform designed specifically for the logistics requirements of Maharashtra. The platform shows improvements in recommendation quality, speed and usability over the traditional manual recommendation process. The platform which integrates machine learning algorithms with LLM capabilities in stable architecture, provides personalized and warehouse recommendations.

The hybrid recommendation strategy, which integrates K-Nearest Neighbors and Random Forest algorithms using meta-learning strategies, indicates better ranking stability and easier to understand than direct algorithm implementations. The fallback strategy ensures that the service is always up and running, while the explanation module is important for building user trust and helping users make better decisions. Future work will involve integrating offline learning capabilities that enable the system to change its weights according to user interactions over time and improving data checking. There is also potential for expanding the system’s coverage to other regions, adding real-time market data and adding features to make the platform more useful for warehouse managers and logistics professionals.

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